

Inferential Statistics

Discrete Distributions

CSDA 5110

Personality preference types: Introvert or Extrovert?

Isabel Briggs Meyers was a pioneer in the study of personality types. In the book *A guide to the Development and Use of the Myers-Briggs Type Indicators*, Myers and McCaully, it was reported that based on a very large sample (2282 professors), approximately 45% of all university professors are extrovert.

Suppose you have classes with 10 different professors.

- What is the probability that exactly 6 are extroverts?
- What is the probability that none of your professors is an extrovert?
- What is the probability that at least 3 of your professors are extrovert?
- What is the *expected number* of extroverts? What is the *standard deviation* of the distribution?

The Binomial Distribution

The **binomial experiment** is a probability experiment that satisfies these requirements:

1. There must be a fixed number of trials.
2. Each trial can have only two outcomes
3. The outcomes of each trial must be independent of each other.
4. The probability of success must remain the same for each trial.

Notation for the Binomial Distribution

$P(S)$ The symbol for the probability of success

$P(F)$ The symbol for the probability of failure

p The numerical probability of success

q The numerical probability of failure

$P(S) = p$ and $P(F) = 1 - p = q$

n The number of trials

X The number of successes in n trials

Note that $X = 0, 1, 2, 3, \dots, n$

Probabilities for a Binomial Distribution

- Denote the probability of success on a trial by p .
- For n independent trials, the probability of x successes equals:

$$P(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

```
> rbinom(n=1, size=10, prob=.4)
```

```
[1] 1
```

Binomial Distribution

- Generating random numbers from the binomial distribution is not simply generating random numbers but rather generating the number of successes of independent trials.
- To simulate the number of successes out of ten trials with probability 0.4 of success, we run `rbinom` with `n=1` (only one run of the trials), `size=10` (trial size of 10) and `prob=0.4` (probability of success is 0.4).

Binomial Distribution

- Similar to the normal distribution functions, `dbinom` and `pbinom` provide the density (probability of an exact value) and distribution (cumulative probability), respectively, for the binomial distribution

```
> # probability of 3 successes out of 10  
> dbinom(x=3, size=10, prob=.3)
```

```
[1] 0.2668279
```

```
> # probability of 3 or fewer successes out of 10  
> pbinom(q=3, size=10, prob=.3)
```

```
[1] 0.6496107
```

```
> # both functions can be vectorized  
> dbinom(x=1:10, size=10, prob=.3)
```

```
[1] 0.1210608210 0.2334744405 0.2668279320 0.2001209490 0.1029193452  
[6] 0.0367569090 0.0090016920 0.0014467005 0.0001377810 0.0000059049
```

```
> pbinom(q=1:10, size=10, prob=.3)
```

```
[1] 0.1493083 0.3827828 0.6496107 0.8497317 0.9526510 0.9894079  
[7] 0.9984096 0.9998563 0.9999941 1.0000000
```

Personality preference types: Introvert or Extrovert?

Isabel Briggs Meyers was a pioneer in the study of personality types. In the book *A guide to the Development and Use of the Myers-Briggs Type Indicators*, Myers and McCaully, it was reported that based on a very large sample (2282 professors), approximately 45% of all university professors are extrovert.

Suppose you have classes with 10 different professors.

- What is the probability that exactly 6 are extroverts?
- What is the probability that none of your professors is an extrovert?
- What is the probability that at least 3 of your professors are extrovert?
- What is the *expected number* of extroverts? What is the *standard deviation* of the distribution?

Example : Survey on Employment

A survey from Teenage Research Unlimited (Northbrook, Illinois) found that 30% of teenage consumers receive their spending money from part-time jobs. If 5 teenagers are selected at random, find the probability that at least 3 of them will have part-time jobs.

The Binomial Distribution

The mean, variance, and standard deviation of a variable that has the *binomial distribution* can be found by using the following formulas.

Mean: $\mu = np$

Variance: $\sigma^2 = npq$

Standard Deviation: $\sigma = \sqrt{npq}$

Example: Unusual Result?

According to CTIA, 25% of ALL US households are wireless-only households (no landline).

In a random sample of 300 households, 92 were wireless-only. Is this result unusual?

Poisson Distribution

The probability of X occurrences in an interval of time, volume, area, etc., for a variable, where λ (Greek letter lambda) is the mean number of occurrences per unit (time, volume, area, etc.), is

$$P(X; \lambda) = \frac{e^{-\lambda} \lambda^X}{X!} \quad \text{where } X = 0, 1, 2, \dots$$

The letter e is a constant approximately equal to 2.7183.

Poisson Distribution

- The **Poisson distribution** is a distribution useful when n is large and p is small and when the independent variables occur over a period of time.
- The Poisson distribution can also be used when a density of items is distributed over a given area or volume, such as the number of plants growing per acre or the number of defects in a given length of videotape.

Poisson Distribution

- Named after French mathematician Simeon Denis Poisson
- Poisson Distribution is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event
- To generate random counts, the density, the distribution and quantiles use **rpois**, **dpois**, **ppois** and **qpois**, respectively.
- As λ (read as Lambda) grows large the Poisson distribution begins to resemble the normal distribution

Example : Typographical Errors

If there are 200 typographical errors randomly distributed in a 500-page manuscript, find the probability that a given page contains exactly 3 errors.

First, find the mean number λ of errors. With 200 errors distributed over 500 pages, each page has an average of

$$\lambda = \frac{200}{500} = 0.4 \quad \text{errors per page.}$$

$$P(X; \lambda) = \frac{e^{-\lambda} \lambda^X}{X!} = \frac{e^{-0.4} (0.4)^3}{3!} = \boxed{0.0072}$$

Thus, there is less than 1% chance that any given page will contain exactly 3 errors.

Example

A McDonald's manager knows that cars arrive at the drive-through at the average rate of two cars per minute between the hours of 12 noon and 1:00pm. She needs to determine and interpret the probability of the following events:

- i. Exactly 6 cars arrive between 12 noon and 12:05 pm.
- ii. Fewer than 6 cars arrive between 12 noon and 12:05pm.
- iii. At least 6 cars arrive between 12 noon and 12:05pm.